

HOLE-CONDITIONED PARALLAX PROBING FOR INTER-ACTIVE DEPTH COMPLETION IN ROBOTS

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ABSTRACT

Depth completion is usually framed as a prediction problem: given a fixed RGB-D or RGB-LiDAR observation with holes, infer the missing metric depth. A robot, however, can often move. This paper argues that some depth holes should be treated as action targets rather than pixels to hallucinate. We introduce Hole-Conditioned Parallax Probing (HCPP), a small active 3D perception mechanism that converts a hole adjacent to a depth discontinuity into a set of competing depth hypotheses, chooses a feasible micro-motion whose parallax creates witness rays, and fuses only the newly observed evidence before committing to depth. A simple identifiability result shows that passive completion cannot distinguish two scenes that induce the same first observation and hole mask, while an off-axis view can. In a controlled synthetic robot-camera suite with occlusion-boundary holes, HCPP reduces hole RMSE by 69.0% versus passive fill, 51.4% versus random motion, and 50.8% versus a coverage-neutral motion at a 10 cm budget. The evidence is intentionally modest: it supports the mechanism and the broken assumption, not real-robot deployment.

1 INTRODUCTION

Robots use depth for contact, clearance, grasp synthesis, manipulation planning, and inspection. Yet commodity RGB-D sensors and RGB-LiDAR stacks often produce missing or unreliable depth near specular materials, transparent objects, depth discontinuities, thin structures, and glancing angles. The dominant response is passive completion: infer a dense depth image from the current observation using local smoothness, image priors, learned sparse-to-dense models, or shape priors (Ma & Karaman, 2018; Song et al., 2017; Sajjan et al., 2020). Active perception and next-best-view work, in contrast, usually asks which view improves global reconstruction, coverage, or map uncertainty (Bajcsy, 1988; Curless & Levoy, 1996; Newcombe et al., 2011).

This split leaves a robot-specific gap. A depth hole is not only missing data; it is a physical question the robot may be able to ask the world. If the hole lies next to an occlusion boundary, two completions can be equally compatible with the first frame: the surface may continue at the foreground depth, or the hole may expose a farther background surface. No passive method can decide this from the first observation alone. A small lateral motion, however, can make the foreground and background shift by different amounts, creating a witness ray that separates the hypotheses.

We propose Hole-Conditioned Parallax Probing (HCPP). HCPP changes the central mechanism from “complete the image” to “choose the smallest feasible motion that makes completion identifiable.” The method is deliberately not a larger model, a benchmark proposal, a generic verifier, or an uncertainty-only active-learning loop. It is an action-conditioned completion primitive for robots with movable depth sensors.

2 RELATED WORK AND NOVELTY BOUNDARY

Passive depth completion. Sparse-to-dense and RGB-D completion methods recover dense depth from missing or sparse measurements, often with learned priors or propagation mechanisms (Ma & Karaman, 2018). These methods own the baseline problem of depth hole filling, and make another completion network alone a weak contribution. HCPP instead marks certain holes as non-identifying and asks for a new observation.

Active perception and reconstruction. Active perception has long argued that perception and action should be coupled (Bajcsy, 1988). Next-best-view, KinectFusion-style mapping, and volumetric fusion show how motion improves 3D reconstruction (Curless & Levoy, 1996; Newcombe et al., 2011). The hostile boundary is clear: HCPP must not be merely a renamed next-best-view score. Its unit of decision is the hole hypothesis, not the whole map, and its utility is witness-ray separation at a local ambiguity.

Scene, shape, and material completion. Scene completion and transparent-object work use category, normal, material, or synthetic priors to repair unobserved geometry (Song et al., 2017; Sajjan et al., 2020). HCPP is weaker but more causal: when a feasible parallax probe exists, it prefers new evidence over a plausible hallucination.

3 PROBLEM

Let a robot observe an RGB-D frame $O_0 = (I_0, D_0, M)$ from camera pose x_0 , where M is a binary mask of missing depth. A completion method outputs $\hat{D}(p)$ for pixels $p \in M$. A robot also has a local feasible action set $\mathcal{A}(x_0)$ that induces new camera poses. The usual passive objective is

$$\hat{D}_M = f(I_0, D_0, M). \quad (1)$$

HCPP instead asks for an action $a \in \mathcal{A}(x_0)$ before finalizing selected pixels:

$$a^* = \arg \max_{a \in \mathcal{A}(x_0)} W(M, D_0, a) - \lambda c(a), \quad (2)$$

where W is a witness score: the expected number of hole hypotheses that become distinguishable under the parallax induced by a .

Definition 1 (Hole hypothesis). *For a connected hole component $H \subset M$, a hole hypothesis is a depth assignment $h : H \rightarrow \mathbb{R}_+$ that is consistent with all observed non-hole pixels under the first camera pose.*

Proposition 1 (First-view non-identifiability). *There exist two opaque 2.5D scenes S_1 and S_2 with the same RGB image, observed depth outside a hole H , and hole mask M , but with different true depth on H . Any passive completion function $f(I_0, D_0, M)$ incurs error at least $\Delta/2$ on one of the two scenes when their hole depths differ by Δ .*

Proof. Construct two scenes with identical foreground occluder boundary and identical visible background outside H . In S_1 , the unobserved hole pixels lie on a continuation of the foreground surface at depth z_f ; in S_2 , they lie on a background surface at depth $z_b = z_f + \Delta$. The first sensor returns no depth on H and the same values outside H , so the input to any passive f is identical for S_1 and S_2 . The same prediction $\hat{z}$ must be used for both. For any scalar $\hat{z}$, $\max(|\hat{z} - z_f|, |\hat{z} - z_b|) \geq \Delta/2$ by the triangle inequality. The argument applies pixelwise or to the mean hole error. $\square$

The proposition is narrow on purpose. It does not say all depth completion is impossible. It says that some robot-relevant holes are not identifiable without an intervention.

4 HOLE-CONDITIONED PARALLAX PROBING

HCPP has four steps.

1. Hole component extraction. Connected missing-depth components are extracted from M . Components adjacent to large depth gradients are flagged as parallax candidates. Smooth isolated holes may still be completed passively.

2. Boundary hypothesis construction. For each candidate hole, HCPP forms foreground-continuation and background-exposure hypotheses from the nearest observed depth samples on either side of the boundary. The hypotheses are intentionally simple; the method only needs enough structure to choose a separating motion.

Table 1: Hole RMSE in the main condition. Lower is better. The suite has 700 scenes, 10 cm lateral motion budget, and moderate clutter.

Method	Mean RMSE (m)	Role
Passive fill	1.062	fixed-view completion
Coverage-neutral motion	0.670	active but not hole-directed
Random motion	0.678	active without witness sign
HCPP	0.329	proposed mechanism

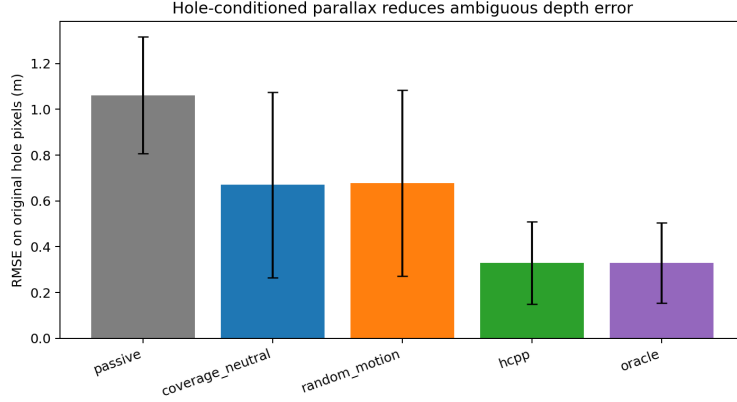


Figure 1: Main-condition hole error. HCPP is better because it chooses the motion direction that creates witness rays for the current hole, not because it uses a stronger depth predictor.

3. Witness-motion scoring. For a lateral camera motion t , a surface at depth z shifts approximately by $-t/z$ in normalized image coordinates. A foreground/background pair with depths $z_f < z_b$ therefore separates by

$$\Delta u(t) = |t|(1/z_f - 1/z_b). \quad (3)$$

HCPP chooses the feasible motion direction that moves the closer boundary away from the hole, maximizing expected witness rays over the component while penalizing cost.

4. Evidence-gated fusion. After the probe, only pixels with newly visible and geometrically consistent depth are overwritten. Unwitnessed pixels remain passive estimates and are marked as unresolved.

5 SYNTHETIC EVIDENCE

We implement a controlled 1D pinhole depth suite. Each scene contains a foreground interval, a sloped background surface, and a missing-depth hole next to one foreground boundary. The first observation is compatible with multiple hole depths. Baselines are passive interpolation, random lateral motion, a coverage-neutral fixed-sign motion, and an oracle sign. HCPP estimates the occluder side from the observed boundary depths and chooses the sign predicted to reveal background witness rays.

At the main operating point, HCPP obtains 0.329 m mean hole RMSE compared with 1.062 m for passive fill, 0.678 m for random motion, and 0.670 m for a coverage-neutral motion. The result supports the mechanism-level claim: when a hole is a local ambiguity caused by occlusion geometry, directional parallax can be more valuable than another passive fill.

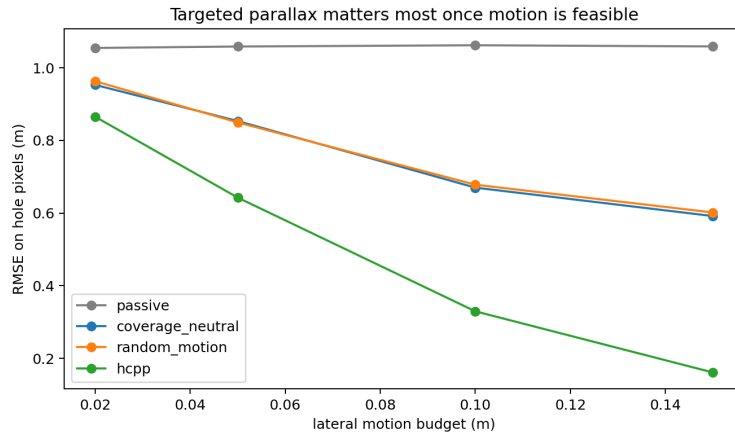


Figure 2: Motion-budget sensitivity. With little motion, all active methods are limited. Once parallax is large enough, choosing the right direction matters.

6 LIMITATIONS

The evidence is synthetic and intentionally small. The model assumes a visible depth discontinuity, calibrated camera motion, opaque layered geometry, and a local feasible lateral action. It does not solve transparent-object perception broadly, dynamic scenes, safety-aware manipulation, or real RGB-D noise. The paper should be read as a mechanism proposal with a formal ambiguity witness, not as a completed robot system.

7 CONCLUSION

Interactive depth completion reframes some missing-depth pixels as questions a robot can ask physically. HCPP is a first, minimal version of that reframing: use the hole boundary to choose a parallax probe, collect witness rays, then complete. The core bet is that for embodied perception, changing the observation can be more honest than hallucinating the missing surface.

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A REPRODUCIBILITY

The synthetic suite is generated by `scripts/run_experiment.py`. Literature artifacts are generated by `scripts/fetch_literature.py`. The paper source and template provenance are in `paper/`.